SCI-INTEL: Acquisition-Aware Retrieval and Quality-Risk Curation of Scientific Microscopy Images

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***Abstract*—Retrieval and quality control of scanning electron microscopy (SEM) archives are complicated because one specimen looks different under different instruments, detectors, and accelerating voltages. We present SCI-INTEL, a local platform that keeps two separate representations: a frozen DINOv2 ViT-S/14 for image-derived quality screening and an acquisition-aware supervised-contrastive adapter for same-specimen retrieval. The branches are combined by deterministic evidence retrieval, rule-based suggested actions, uncertainty-aware abstention, and curator review, not by learned fusion. On an instrument-held-out split of the HCCI collection (212 queries; three specimens shared across splits), the adapter reduced the observed acquisition-geometry similarity gap from 0.2016 to 0.0681 (66.23%; paired dz = 2.19, p = 5.03×10^-36, 210 queries) and raised Recall@5 from 0.9858 to 0.9921 under an unmasked-distractor protocol, but lowered 11-class synthetic-artifact macro-F1 from 0.6837 to 0.6323. Frozen DINOv2 reached binary AUROC 0.8582, and model-derived suspicious-region localization reached macro IoU 0.4454 (Dice 0.5103) on 500 synthetic masks. Selective prediction reached 99.08% accuracy at 29.73% coverage, with weak calibration (ECE 0.3333), and evidence was available for all 55 evaluated queries. Results rest on synthetic artifacts and limited specimen coverage; they do not establish physical-defect detection or unseen-specimen generalization.**

***Index Terms*—scientific microscopy, image retrieval, acquisition robustness, quality assessment, human-in-the-loop curation**

I. INTRODUCTION

Scanning electron microscopy (SEM) images depend strongly on how they are acquired. Accelerating voltage, detector (secondary or backscattered electrons), working distance, and focus change contrast and apparent texture, so one specimen can look more different across settings than different specimens do under identical settings. In shared archives, similarity search then returns same-acquisition neighbors instead of same-specimen evidence, and quality control is slow and subjective.

Four problems follow: (1) acquisition-induced visual variance; (2) the unverified benefit of acquisition metadata for ranking; (3) redundant re-scans, where aggressive deduplication could discard legitimate acquisitions; and (4) automated suggestions without a human decision point or traceable records. We treat the task as scientific-image retrieval and quality-risk curation. It is not diagnosis, and every indicator is image-derived and advisory.

Contributions: (C1) Acquisition-aware retrieval: a supervised-contrastive adapter on frozen DINOv2 features, trained with same-acquisition pairs masked and evaluated with unmasked distractors on an instrument-held-out split. (C2) A quantified specialization trade-off: adaptation helps retrieval but lowers artifact sensitivity, so screening stays on the frozen representation. (C3) Screening and evidence: image-derived quality risk, model-derived suspicious-region localization, evidence retrieval, and uncertainty-aware abstention, evaluated on controlled synthetic degradations. (C4) Engineering integration: a reproducible local platform that persists curator decisions, provenance events, and audit logs. C1 and C2 are methodological; C4 is integration.

II. RELATED WORK

Micrograph retrieval: Handcrafted descriptors such as GLCM textures [1], SIFT [2], and visual-word signatures [3] were used for microstructure search; see [4] and CBIR [5]. MicroNet pre-training improves micrograph segmentation [6].
Foundation features: Vision Transformers [7] trained by self-distillation [8], masked modeling [9], or DINOv2 [10] provide general features; ResNet [11] remains a common baseline.
Acquisition shift: Contrastive [12] and supervised contrastive [13] objectives shape metric spaces, and domain-divergence theory [14] motivates treating instrument change as distribution shift.
Indexing: FAISS [15] and HNSW [16] provide vector search.
Quality & abstention: No-reference quality [17], structural similarity [18], anomaly benchmarks [19], selective classification [20, 21], and calibration [22] inform our screening and abstention.
Curation: Perceptual hashing [23] and FAIR principles [24] inform deduplication and provenance.

III. SCI-INTEL ARCHITECTURE AND METHOD

Fig. 1 shows the workflow: upload, validation, storage, metadata, representation, quality screening, localization, retrieval, evidence retrieval, explanation, suggested action, uncertainty handling, review, curation, audit, and provenance. A FastAPI backend with JWT authentication, an SQLAlchemy database, FAISS indices, and a React workbench implement it locally. Two representations stay separate: dinov2_base (frozen DINOv2 ViT-S/14) and phase4_adapted (adapter head on DINOv2 features). They share no weights and are combined only by deterministic orchestration.

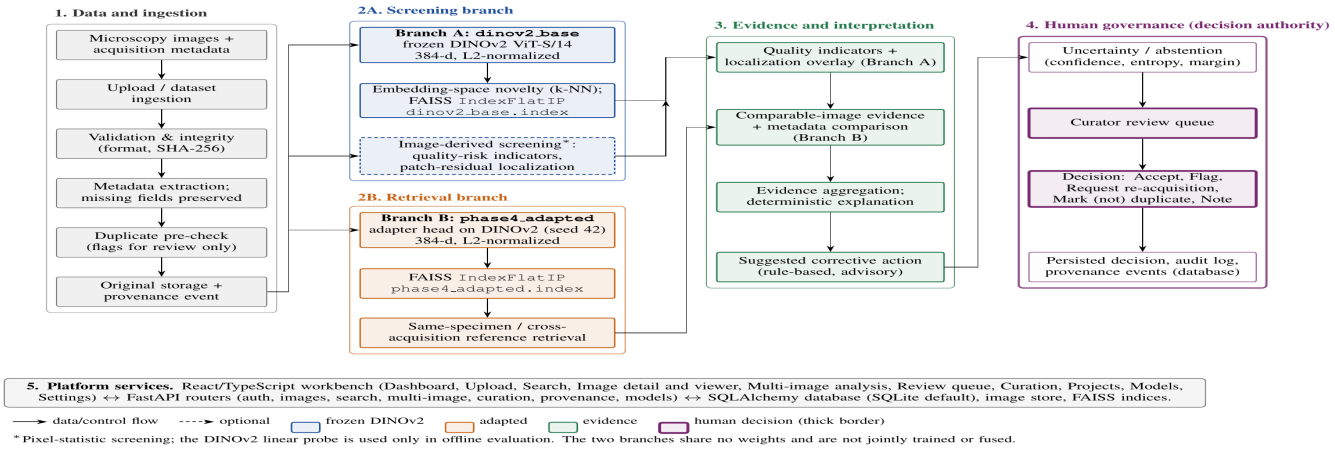


Fig. 1. SCI-INTEL architecture. The branches dinov2_base and phase4_adapted share no weights and are not jointly trained or fused; they are combined only by deterministic orchestration. Curators remain the decision authority.

Representation & retrieval: Images are normalized and encoded into 384-d vectors, L2-normalized: ■ = z / ■z■■, s(q, x) = ■_q■ ■_x (1). The adapter maps normalized features to a unit-norm 384-d embedding. Positives P(i) are same-specimen, different-acquisition pairs; same-specimen same-acquisition peers are excluded (2). AdamW [25] trained 427 images with validation selection on 135 images. Gap Δ = s■_within – s■_cross (3).

Screening, evidence & review: Quality risk combines Laplacian variance, entropy, clipping, and FFT high-frequency ratio (flag ≥ 0.50). Localization thresholds a 16×16 patch-residual saliency map (|patch mean − image median| ≥ 0.50). Novelty evaluates k-NN cosine distance. Uncertainty abstains at confidence < 0.40, entropy > 0.75, or margin < 0.10. Evidence retrieval returns comparable references and advisory actions.

IV. EXPERIMENTAL SETUP

Table I lists the 6,085 active micrographs. The HCCI split [27] (the record lists 777 micrographs; the deposited archive holds 774) is instrument-disjoint (train: Helios; validation: VEGA3; test: Zeiss Gemini) with no image overlap, but three steel specimens occur in every split. The task is acquisition-robust same-specimen retrieval, not unseen-specimen generalization. Carinthia [28] (100 images) serves a cross-domain screen; BBBC021 [26] profiles ingestion. Larger registered collections [29] were not evaluated. Protocol U tests 212 queries against 211 candidates each.

Dataset	Images	Format	License	Role
HCCI (SEM, cast iron)	774	PNG	CC BY 4.0	Retrieval; synthetic
Carinthia (SEM)	4,591	JPEG	CC BY 4.0	Novelty screen
BBBC021 (bioimage)	720	TIFF (16-bit)	CC0 1.0	Metadata profiling
Total active micrographs	6,085	Multi-format	–	Evaluated repository cohort
HCCI Train	427	PNG	CC BY 4.0	Helios NanoLab, Helios G4 PFIB CXe
HCCI Validation	135	PNG	CC BY 4.0	VEGA3 XMH
HCCI Test (queries)	212	PNG	CC BY 4.0	Zeiss Gemini (211 candidates/query)

TABLE I: Datasets and HCCI Instrument-Disjoint Partition (3 Shared Steel Specimens).

V. RESULTS

A. Acquisition gap and retrieval: The adapter shrank the observed acquisition gap from 0.2016 (within 0.7811, cross 0.5794) to 0.0681 (within 0.9085, cross 0.8404), a reduction of 66.23% (95% CI 65.72–67.19; query-level 66.40%; dz = 2.19, p = 5.03×10^-36 over 210 queries; seeds: 60.76%, 70.35%, 67.57%; Fig. 2a). Under Protocol U (Table II, Fig. 2b) the 3-seed adapter attains R@1 0.1447, R@5 0.9921, and MRR 0.5261, versus 0.1321, 0.9858, and 0.5200 for frozen DINOv2 (intervals overlap; gap reduction is statistically strong).

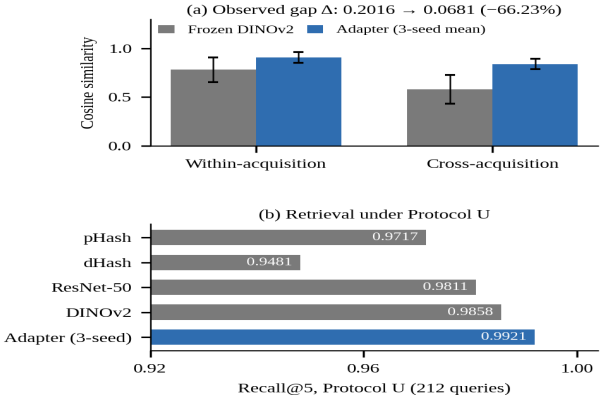


Fig. 2. (a) Observed gap reduction (0.2016 to 0.0681, -66.23%). (b) Recall@5 under Protocol U (212 queries, 211 candidates each).

Method	R@1	R@5	MRR	P@5
pHash-64	0.0991	0.9717	0.4894	0.6736
dHash-64	0.1085	0.9481	0.4995	0.6236
ResNet-50	0.0802	0.9811	0.4836	0.5726
DINOv2 ViT-S/14	0.1321	0.9858	0.5200	0.6160
Adapter, 3-seed mean	0.1447	0.9921	0.5261	0.6327

TABLE II: Protocol U Retrieval Benchmarks (212 Queries, 211 Candidates Each).

B. Quality screening and trade-off: Binary risk metrics (Table III) are AUROC/AUPRC 0.8281/0.9808 (handcrafted), 0.8582/0.9841 (frozen DINOv2), and 0.8230/0.9792 (adapted). On 11-class classification, macro-F1 falls from 0.6837 (balanced acc 0.7036) to 0.6323 (0.6645) after adaptation, confirming that adaptation trades artifact sensitivity for retrieval and motivating dual representations.

Metric	Handcrafted	Frozen DINOv2	Adapted (3-seed)
Retrieval R@5 (Protocol U)	—	0.9858	0.9921
Retrieval MRR (Protocol U)	—	0.5200	0.5261
Acquisition gap Δ (lower better)	—	0.2016	0.0681
Risk AUROC (binary)	0.8281	0.8582	0.8230
Risk AUPRC (binary)	0.9808	0.9841	0.9792
11-class macro-F1	—	0.6837	0.6323
11-class balanced accuracy	—	0.7036	0.6645

TABLE III: Representation Specialization Trade-off Matrix.

C. Localization, uncertainty, evidence, latency: Localization (Fig. 3b) reaches macro IoU 0.4454 and Dice 0.5103 on 500 synthetic masks. Selective prediction (Fig. 3a) reaches 99.08% accuracy at 29.73% coverage ($\tau = 0.4$) and 100% at 9.73% coverage ($\tau = 0.6$; ECE 0.3333). For N = 55 queries (275 items; Table IV), evidence availability was 100.0% with 0.0% duplicates. Novelty AUROC was 0.7404 on HCCI and 1.0 on Carinthia.

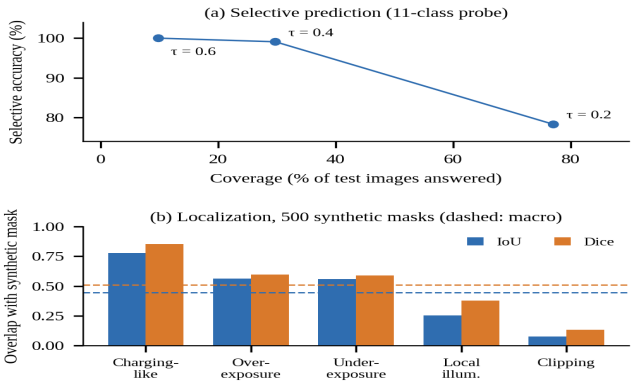


Fig. 3. (a) Coverage vs. selective accuracy at $\tau = 0.2, 0.4, 0.6$. (b) Per-category localization IoU and Dice on 500 synthetic masks.

Evidence Cohort Parameter (N = 55 queries, 275 items)	Measurement Value
Valid evidence availability	100.0%
Same-specimen cross-acquisition	100.0%
Cross-instrument evidence	100.0%
Quality-compatible evidence	100.0%
Duplicate evidence items	0.0%
Missing provenance	0.0%
Deterministic ranking reproducibility	100.0%
Mean evidence items / query	5.0

TABLE IV: Grounded Evidence Cohort Verification (N = 55 Queries, 275 Evidence Items).

Orchestration latency: The sealed Phase-6 log records 46.00 ms mean and 51.05 ms P95 per image on CPU (stages: preprocessing 1.27, representation 0.04, screening 5.44, localization 12.20, retrieval 9.44, aggregation 17.61 ms; timing passes synthetic arrays and excludes encoder inference). An earlier report lists 23.40 ms (P95 28.30 ms); it reflects a different profile and is not end-to-end.

VI. DISCUSSION AND LIMITATIONS

Late score-level metadata fusion gave no gain (validation MRR was flat for $\alpha \geq 0.1$, tie-break selected $\alpha^* = 1.0$, test $\Delta R@1 = 0.0000$); visual features are saturated, but metadata remains essential for relational constraints. Limitations: All splits share 3 alloys; unseen-specimen generalization is not claimed.

Defect labels and masks are synthetic. Localization is a 16-px heuristic. Calibration is weak (ECE 0.3333). Evidence cohort has 55 queries. Actions are untested on physical microscopes. The platform serves one seed-42 head.

VII. CONCLUSION AND FUTURE WORK

SCI-INTEL reduced the observed acquisition-geometry similarity gap by 66.23% on an unseen instrument while the frozen backbone remained the better artifact screener, and it composes the two deterministically with uncertainty-aware human review. Future work includes held-out alloys, expert-labeled defects, cross-attention fusion, and microscopist user studies.

COMPLIANCE WITH ETHICAL STANDARDS

This study utilized publicly available scientific imaging datasets (HCCI bearing steel, Carinthia SEM, and BBBC021 bioimage benchmark) and controlled synthetic augmentations. The research did not involve human subjects, patient data, clinical interventions, or live animal experiments, and therefore did not require institutional review board (IRB) approval.

DISCLOSURES & CONFLICTS OF INTEREST

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